TIPA Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Mineralized Collagen Fibril-Based Biomaterial Platform for *In Vitro* 3D Bone Microenvironment Modeling**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), **Advanced Bio (√),** Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio & Medical		**Bio & Medical**	
	Medium Category	Small Category	Medium Category	Small Category
	Medical Bio	**Bio - Biomaterials**	**Industrial Biotechnology**	**Functional Biomaterials**
Concept of Proposed Technology	The proposed technology is a **mineralized collagen fibril (MCF)-based biomaterial platform** designed to recreate key structural, compositional, and mechanical features of native bone extracellular matrix (ECM) for ***in vitro* 3D bone-like microenvironment modeling**. Unlike conventional collagen or hydroxyapatite composites, the technology focuses on **controlled mineralization at the collagen fibril level**, producing fibrillar architectures that more closely mimic native bone ECM in terms of nanoscale organization, stiffness, and biochemical cues. These MCFs can be incorporated into existing 3D scaffold systems, hydrogels, or cell culture platforms without requiring major changes to scaffold geometry.			

	Potential applications include - 3D cell culture systems for bone biology and disease modeling - Evaluation of osteogenic differentiation and bone-cell interactions - Advanced *in vitro* testing platforms for regenerative medicine and biomaterials research
Rationale for Joint R&D Project	While mineralized collagen-based biomaterials have demonstrated strong potential in academic research, their translation into **scalable, industry-ready platforms** requires close collaboration between **globally recognized research institutions and industrial partners**. A joint R&D project is necessary to - Optimize MCF production for **reproducibility and scalability** - Integrate MCFs into **industry-relevant 3D culture or scaffold platforms** - Validate performance using **standardized *in vitro* assays** aligned with existing SME product pipelines Collaboration with Korean SMEs enables the alignment of advanced biomaterial research with **manufacturing constraints, product requirements, and downstream commercialization pathways**, accelerating the transition from laboratory-scale validation to product-ready technologies. The **University of Toronto (UofT)** provides a uniquely strong environment for such joint R&D through its role as a **designated TIPA overseas centre**, offering Korean SMEs structured access to advanced biomaterials expertise, established translational research infrastructure, and a proven framework for international collaboration. UofT's participation enables Korean SMEs to de-risk early-stage technology development while accelerating alignment with downstream commercialization pathways supported by TIPA. A joint R&D project with UofT is particularly well suited to this technology because it enables - **Advanced materials design and characterization** capabilities at the research-institute level, allowing precise control and validation of mineralized collagen fibril (MCF) structure, composition, and mechanical properties. - **Systematic integration of MCFs into industry-relevant 3D culture or scaffold platforms**, informed by SME product constraints and manufacturing considerations. - **Standardized *in vitro* validation using globally competitive research protocols**, supporting reliable benchmarking and future regulatory or commercial positioning. Furthermore, UofT's established **UT-TIPA Center infrastructure** facilitates efficient proposal development, communication with Korean SMEs, and transition from technology proposal to full R&D project, improving both proposal quality and success probability within the TIPA program framework.

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	Through this collaboration, Korean SMEs gain access to **front-line biomaterials research**, while UofT ensures that technology development is guided by **industry relevance, scalability, and commercialization potential**, enabling a faster and more robust transition from laboratory-scale validation to product-ready innovation.
Key Words	1. Mineralized collagen fibrils 2. Bone microenvironment 3. Functional biomaterials 4. 3D cell culture 5. Biomimetic ECM

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (7)
R&D Objectives	The primary objectives of this joint R&D project are to - Develop a **reproducible and scalable process** for producing mineralized collagen fibrils suitable for industrial application. - Integrate MCFs into existing **3D biomaterial or cell culture platforms** used by SMEs. - Demonstrate that MCF-based materials **enable enhanced bone-like cellular responses** compared to non-mineralized collagen systems. - Validate performance through **quantitative *in vitro* assays** relevant to bone biology and regenerative medicine.			
R&D Target Specifications	Target performance metrics include - **Controlled mineral content**: tunable CaP ratios comparable to native bone ECM - **Mechanical properties**: elastic modulus within the range of trabecular bone-mimetic substrates (e.g. 10-100 MPa at material level) - **Structural fidelity**: preservation of fibrillar collagen architecture after mineralization - **Biological response**: increased osteogenic marker expression (e.g. ALP activity, RUNX2 expression) compared to collagen-only controls - **Compatibility**: ability to integrate MCFs into existing scaffold or hydrogel platforms without altering macroscopic design			
R&D Plan	**Phase 1 (TRL 3–4): Laboratory-scale optimization** - Synthesis and characterization of mineralized collagen fibrils - Structural, chemical, and mechanical analysis - Initial *in vitro* biocompatibility and osteogenic response testing **Phase 2 (TRL 5–6): Prototype integration and validation** - Incorporation of MCFs into SME-relevant 3D culture or scaffold systems - Optimization of processing parameters for reproducibility - Expanded *in vitro* testing using standardized cell assays **Phase 3 (TRL 7): Reliability and application testing** - Evaluation of MCF-based platforms under industry-relevant conditions - Benchmarking against existing biomaterial solutions - Preparation for downstream product development and commercialization			

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	- Design and synthesis of mineralized collagen fibrils - Advanced materials characterization (structural, chemical, mechanical) - *In vitro* biological evaluation and data analysis - Technical guidance on biomimetic ECM design principles - Knowledge transfer and joint publication/patent planning as appropriate
Expected R&R of Korean SME Partner	- Integration of MCFs into existing biomaterial or 3D culture platforms - Scale-up considerations and process optimization - Application-specific testing and validation - Feedback on industrial feasibility and product requirements - Support for sterilization, commercialization strategy and regulatory considerations

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	This joint R&D project is expected to deliver - A **validated MCF-based biomaterial platform** suitable for advanced *in vitro* bone microenvironment modeling - Enhanced differentiation and functional outcomes in bone-related 3D cell culture systems - New opportunities for Korean SMEs to expand product portfolios in **advanced biomaterials and regenerative medicine** - Strengthened international collaboration between Korean industry and global research institutions - A foundation for **future commercialization, licensing, or product development under the TIPA framework**

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization